

This is my first python program

Hello Python

```
In [2]: print("hello world")
```

hello world

Data Types

```
In [3]: x = 3  
print(x)
```

3

```
In [4]: type(x)
```

Out[4]: int

```
In [5]: y = 3.4567788889  
print(y)
```

3.4567788889

```
In [6]: type(y)
```

Out[6]: float

```
In [8]: z = 23+67j  
print(z)  
type(z)
```

(23+67j)

Out[8]: complex

```
In [9]: a = "I am learning python"  
print(a)  
type(a)
```

I am learning python

Out[9]: str

how many variables are there in notebook now?

In [10]: %whos

Variable	Type	Data/Info
a	str	I ma learning python
b	int	16
x	int	3
y	float	3.4567788889
z	complex	(23+67j)

Deleting variable from the memory

In [11]: `del x`

In [12]: %whos

Variable	Type	Data/Info
a	str	I ma learning python
b	int	16
y	float	3.4567788889
z	complex	(23+67j)

Multuple Assignments

In [19]: `a,b,c=12,34.5,"hello"`
`print(a,b,c)`
`print(type(a), type(b),type(c))`

12 34.5 hello
<class 'int'> <class 'float'> <class 'str'>

Case Sensitiive

In [21]: `a= 12`
`A= "student"`
`print(a)`

12

Python Arithmetic Operators

In [23]: %whos

Variable	Type	Data/Info
A	str	student
a	int	12
b	float	34.5
c	str	hello
y	float	3.4567788889
z	complex	(23+67j)

In [24]: `sum = a+y`
`print(sum)`

15.4567788889

In [25]: `print(type(sum))`

<class 'float'>

In [28]: `print(c + A)`

hellostudent

In [30]: `print(c + " " + A)`

hello student

In [31]: `print(z + a)`

(35+67j)

Subtract Operator

In [32]: %whos

Variable	Type	Data/Info
A	str	student
a	int	12
b	float	34.5
c	str	hello
sum	float	15.4567788889
y	float	3.4567788889
z	complex	(23+67j)

In [34]: `print(b-a)`

22.5

```
In [35]: print(z-a)
```

```
(11+67j)
```

```
In [36]: print(sum-y)
```

```
12.0
```

Multiplication Operator

```
In [38]: %whos
```

Variable	Type	Data/Info

A	str	student
a	int	12
b	float	34.5
c	str	hello
sum	float	15.4567788889
y	float	3.4567788889
z	complex	(23+67j)

```
In [39]: print(a*b)
```

```
414.0
```

```
In [40]: print(a*z)
```

```
(276+804j)
```

```
In [45]: print(sum * y)
```

```
53.43066695354472
```

Divison Opertor

```
In [46]: %whos
```

Variable	Type	Data/Info

A	str	student
a	int	12
b	float	34.5
c	str	hello
sum	float	15.4567788889
y	float	3.4567788889
z	complex	(23+67j)

```
In [47]: print(a/b)
```

```
0.34782608695652173
```

```
In [48]: print ( 36/12)
```

```
3.0
```

```
In [49]: print(z/a)
```

```
(1.9166666666666667+5.583333333333333j)
```

```
In [52]: print(sum/y)
```

```
4.471439853596937
```

Modulus Operator

```
In [54]: n=3
```

```
In [55]: %whos
```

Variable	Type	Data/Info
A	str	student
a	int	12
b	float	34.5
c	str	hello
n	int	3
sum	float	15.4567788889
y	float	3.4567788889
z	complex	(23+67j)

```
In [56]: print(a%n)
```

```
0
```

```
In [57]: print(a%y)
```

```
1.6296633332999995
```

```
In [58]: print(14%5)
```

```
4
```

```
In [62]: print(b%a)
```

```
10.5
```

Exponent Operator

In [63]: %whos

Variable	Type	Data/Info
A	str	student
a	int	12
b	float	34.5
c	str	hello
n	int	3
sum	float	15.4567788889
y	float	3.4567788889
z	complex	(23+67j)

In [64]: print(a**n)

1728

In [65]: print(5**2)

25

In [67]: print(b**n)

41063.625

In [68]: print(z**2)

(-3960+3082j)

In [69]: print (3**2 + 5**2)

34

In [70]: print((25**2) + 35 - (24*5) +(99/3))

573.0

Floor Divison

In [71]: %whos

Variable	Type	Data/Info
A	str	student
a	int	12
b	float	34.5
c	str	hello
n	int	3
sum	float	15.4567788889
y	float	3.4567788889
z	complex	(23+67j)

```
In [72]: print(a//n)
```

4

```
In [74]: print(sum//a)
```

1.0

Can a variable name start with a digit?

```
In [1]: 12x = 23
```

```
File "C:\Users\lenovo\AppData\Local\Temp\ipykernel_14452\3537202119.py", line 1
    12x = 23
      ^
SyntaxError: invalid syntax
```

```
In [8]: _a=6
```

```
In [9]: print(_a)
```

6

python assignment operator

```
In [12]: m = 15
```

```
In [13]: m+=10
print(m)
```

25

```
In [14]: m-=10
print(m)
```

15

```
In [15]: m*=10
print(m)
```

150

```
In [17]: m/=10
print(m)
```

1.5

```
In [26]: m=15
```

```
In [27]: m**=2
```

```
In [28]: print(m)
```

225

python Logical & Boolean Operators

```
In [34]: a= True  
b= True  
c= False  
d = False
```

```
In [31]: print ( a and b )
```

True

```
In [32]: print( a and c)
```

False

```
In [33]: print( c and a )
```

False

```
In [35]: print( c and d)
```

False

```
In [36]: print ( a or b)
```

True

```
In [40]: print ( a or c)
```

True

```
In [41]: print( c or a )
```

True

```
In [42]: print ( c or d)
```

False


```
In [43]: print ( not a)
```

False

```
In [44]: print ( not c)
```

True

```
In [45]: print( ( a and b) or print (c and d) or ( not a))
```

True

comparison Operators

```
In [47]: print ( 2 <3 )
```

True

```
In [49]: c = 2 < 3  
print ( type (c))
```

<class 'bool'>

```
In [50]: d = 2==3  
print (d)
```

False

```
In [51]: 3 == 3.0
```

Out[51]: True

```
In [52]: p = 12  
q = 8  
r = -3  
s = 5.56
```

```
In [53]: print((q<p) and (r>s))
```

False

```
In [54]: print (not((p==q) or (r==s)))
```

True

```
In [55]: True or False and True
```

Out[55]: True

```
In [58]: False or True and False
```

```
Out[58]: False
```

```
In [59]: False and False or True
```

```
Out[59]: True
```

```
In [60]: print((not(2!=3) and True) or (False and True))
```

```
False
```

Bitwise Operator in Python

```
In [61]: a = 5  
b = 4  
print ( a & b)
```

```
4
```

```
In [62]: print(type(a & b))
```

```
<class 'int'>
```

```
In [63]: print ( a | b)
```

```
5
```

```
In [65]: print( a ^ b)
```

```
1
```

```
In [66]: print(~a)
```

```
-6
```

```
In [68]: print(~b)
```

```
-5
```

```
In [69]: print(a <<2)
```

```
20
```

```
In [70]: print(a >>2)
```

```
1
```

In []: